ATOM COUNT CARD: BAKING SODA

ACTIVITY 2

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BSCPE-1A

500g of Sodium Bicarbonate ($NAHCO_3$)



A PRODUCT IDENTITY

- **Product : 500g pure baking soda, (17.64 oz):**
- **Compound : Baking soda (sodium bicarbonate) NAHCO3**
- **Why it matters : Serves as a safe, highly visual, and affordable tool for teaching scientific concept.**

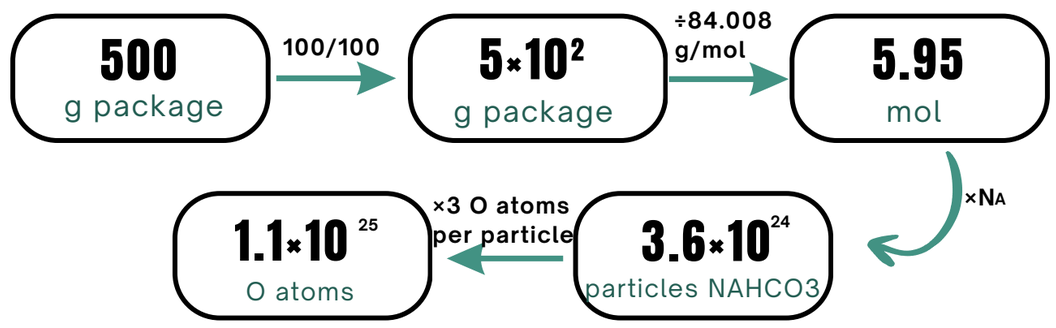
B BASE QUANTITAVE DATA

Quantity	Value	Source	Precision
Net content	500g	Product label	Not Limiting
Atomic masses NA, H, C, O	22.99, 1.008, 12.001, 15.999	PubChem(2006)	Not Limiting
Avogadro's Constant	$6.022\ 140\ 76 \times 10^{23}$	BIPM(2019)	Exact
Formula $NAHCO_3$	subscripts 1, 1, 1, 3	PubChem(2026)	Exact counts

C MOLAR MASS BUILD-UP

Element	Atoms	g/mol each	g/mol	% by mass
NA	1	22.99	22.99	27.3664
H	1	1.008	1.008	1.1998
C	1	12.01	12.01	14.2962
O	3	15.999	47.997	57.1338
NAHCO3	**6**		**84.008**	**100.000**

D THE MOLE MAP: 1 500g BOX



E BASE QUANTITAVE DATA

1. Molar Mass of NAHCO3

$1(22.99) + 1(1.008) + 1(12.01) + 3(15.999) =$ **84.008 g/mol**

2. Label Bridge

$$500\ g \times \frac{100g}{100g\ product} = 500\ g \rightarrow \textbf{5×10² g}$$

3. Mass → Moles

$$500\ g \times \frac{1\ mol}{84.008\ g} = 5.9518\ mol \rightarrow \textbf{5.95 mol}$$

4. Moles → Particles

$$5.9518\ mol \times \frac{6.022\ 140\ 76 \times 10^{23}}{1\ mol} = 3.5842\times10^{24} \rightarrow \textbf{3.5×10}^{\textbf{24}}\textbf{ particles}$$

5. Atoms of one element inside the formula

$$3.5842\times10^{24}\ particles \times \frac{3\ O\ atoms}{1\ particle} = 1.0752\times10^{25} \rightarrow \textbf{1.1×10}^{\textbf{25}}\textbf{ O atoms}$$

6. Percent Composition + Two-route Check (sodium)

$$\%NA = \frac{22.99\ g/mol}{84.008\ g/mol} \times 100\% = 27.366\ \%$$

Route A: $500g \times 0.27366 = 136.83\ g$

Route B: $5.9518\ mol \times \frac{1\ mol\ NA}{1\ mol} \times \frac{22.99\ g}{1\ mol\ NA} = 136.83\ g$

Both Routes: → **1.4×10²**

7. Reverse: from % composition to the formula

In 500 g: NA 27.366 ÷ 22.99 = 1.1903 mol; H 1.998 ÷ 1.008 = 1.1902 mol; C 14.2962 ÷ 12.01 = 1.1903 mol; O 57.1338 ÷ 15.999 = 3.5710 mol

÷ 1.1902 → 1.000: 1.000: 1.000: 3.000 → **NAHCO 3**

8. Reverse: how much particle is one mole

$$1\ mol \times \frac{84.008\ g}{1\ mol} = 84.008\ g$$

$$package\ per\ mole = \frac{84.008\ g}{500g\ package} = 0.168 \rightarrow \textbf{16.8\% package}$$

F SCALE-UP: FROM ONE PACKAGE TO THE WHOLE SCHOOL

1. One box
500 g of baking soda
5.95 mol
3.5×10^{24} particles

2. One lab, One year
Assume 20 boxes a year
1.2×10^3 mol
$20 \times 5.9518\ mol = 119.036\ mol$

3. School, One year
Assume 10 labs
1.2×10^4 mol
$200 \times 5.9518 = 1190.36$

4. Reverse
one mole of NAHCO3 is about
0.168/16.8%
of the package

F SIGNIFICANT-FIGURE CHECK

Calculator: 3.5842×10^{24} molecules
Reported: 3.5×10^{24} molecules

- "500g" has ambigous zeros, as a nominal mass it does not set the precision.

ONE NUMBER TAKEAWAY

One mole of sodium bicarbonate, 6.022×10^{23} particles, fits only about 0.168, about 16.8% of the box.

AI DISCLOSURE

Chat-gpt was used to check calculation and organizing information. All the content and calculation was verified by the student.

REFERENCES:

National Center for Biotechnology Information (2026). PubChem Compound Summary for CID 516892, Sodium Bicarbonate. Retrieved October 13, 2026 from https://pubchem.ncbi.nlm.nih.gov/compound/Sodium-Bicarbonate.

Bureau International des Poids et Measure. (2019). The International Systems of Units (SI) (9th ed). https://www.bipm.org/en/publication/si-brochure